

EFFICIENT BAYESIAN INFERENCE FROM NOISY PAIRWISE COMPARISONS

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ABSTRACT

Assessing the quality of machine-generated content is challenging across essentially every generative modality, because automatic metrics frequently fail to track what humans actually prefer. Human evaluation is a more faithful gold standard, but it is costly and noisy: participants differ widely in expertise, attention, and diligence, whether they are domain experts or crowd workers. Pairwise comparisons improve consistency and transfer readily across tasks and domains, yet turning them into overall quality scores demands careful modeling. Bradley-Terry-based methods update item scores from such comparisons and are used throughout the sciences and machine learning, but existing approaches either ignore rater variability or lack convergence guarantees, which limits their robustness and interpretability in practice. We introduce BBQ, a general Bayesian Bradley-Terry variant that explicitly models rater quality, downweighting or removing unreliable participants, and provides guaranteed monotonic likelihood convergence through an Expectation-Maximization algorithm. Across a broad range of human-evaluation datasets spanning both language and vision, BBQ achieves faster convergence, well-calibrated uncertainty estimates, and more robust, interpretable rankings than baseline Bradley-Terry models, and it remains reliable even under noisy or crowdsourced conditions. More broadly, the framework offers a reliable and cost-effective route to human evaluation wherever noisy pairwise judgments must be aggregated into trustworthy rankings.

1 INTRODUCTION

Evaluating generative models is a challenge that cuts across modalities, from large language models (LLMs) to image generators and beyond, because standard automatic metrics often fail to reflect human preferences. This gap is well documented for the metrics most commonly used in each domain: BLEU (Papineni et al., 2002) and perplexity (Jelinek, 1998) for LLMs, and PSNR (Gonzalez, 2009), MS-SSIM (Wang et al., 2003), and FID (Heusel et al., 2017) for image models all provide only limited insight into perceived quality (Mentzer et al., 2020; CLIC, 2025; Chiang et al., 2024), and analogous limitations recur for other generative tasks. As a result, human evaluation remains indispensable for establishing meaningful rankings between models. Accordingly, leaderboards and comparative studies across fields as varied as large language models and Learned Image Compression (LIC) place strong emphasis on human preference data, and the same reliance is emerging wherever generative systems are compared.

However, human evaluation in any domain is expensive and time-consuming, making it crucial to design protocols that are efficient while minimizing subjectivity and noise. Across these settings, pairwise comparisons, in which participants choose between two items rather than providing absolute scores, are a broadly effective and practical form of human evaluation (Zerman et al., 2018; Wang et al., 2023). They are generally easier for participants and produce more consistent judgments, regardless of the type of content being compared. Collecting all pairwise comparisons is infeasible whenever the number of items is nontrivial, because the number of pairs grows quadratically. And with only a limited set of comparisons, simple statistics such as win rates are not sufficient to derive overall quality scores, because a win rate depends on the quality of the items it is compared against.

To produce an overall ranking that can be displayed on a leaderboard or used in downstream analysis, pairwise comparisons must be aggregated effectively. This motivates the development of aggregation methods that are robust and efficient enough to apply broadly. The Bradley-Terry (BT) (Bradley & Terry, 1952) model addresses this problem by iteratively updating item scores based on comparison outcomes. Owing to its generality, it has become a standard approach for aggregating pairwise judgments across a wide range of domains.

Despite its broad usefulness, the standard Bradley-Terry model has several limitations that surface across the settings in which it is applied. BT parameters can be estimated with a variety of algorithms, including iterative scaling methods (Ford Jr, 1957), gradient descent, and the minorization maximization (MM) algorithm (Hunter, 2004). The MM algorithm guarantees convergence to the global maximum for the basic BT model, whereas gradient-based methods, which are closely related to the Elo rating system, are widely used in practice but provide no such guarantees (Hunter, 2004). Once the model is extended to accommodate the factors that arise in real applications, such as home field advantage (Agresti, 2010), multiple comparisons (Plackett, 1975; Luce et al., 1959), or ties (Rao & Kupper, 1967), even MM algorithms may converge only to local optima (Hunter, 2004). A further and pervasive challenge is that human evaluation is inherently noisy: participants may rush, guess, or lose focus, leading to uneven reliability across raters in virtually any study. If such variability is ignored, unreliable raters can distort item scores and reduce ranking stability.

We propose a general Bayesian Bradley-Terry variant that jointly models item quality and rater reliability. To the best of our knowledge, no prior work combines the BT framework with a Bayesian formulation that explicitly models rater quality while providing closed-form EM updates. Previous approaches that account for noisy raters typically rely on gradient-based optimization without convergence guarantees, whereas our method ensures stable convergence and yields interpretable parameters in any application. The Bayesian framework addresses epistemic uncertainty and provides regularization. To capture variability in participant behavior, we introduce rater-specific parameters that reflect how consistent or trustworthy each participant is, allowing the model to adjust the influence of individual comparisons. We derive an EM algorithm that efficiently estimates item and rater parameters via a latent-variable formulation, guaranteeing monotonic likelihood improvement. This approach allows partially reliable raters to contribute meaningful information while reducing the impact of inattentive or inconsistent participants, without any assumptions specific to a particular task or content type. By modeling rater quality and adopting a Bayesian framework, our method also provides uncertainty estimates for item scores, facilitating more interpretable rankings and robust comparisons across studies and settings.

We demonstrate the effectiveness of our approach on a diverse collection of human evaluation datasets spanning both language and vision, and we introduce a newly collected *inhomogeneous rater quality (IHQ)* dataset¹. Across these varied conditions, our method shows faster convergence, improved robustness to noisy comparisons, and more consistent rankings than standard BT and naive aggregation methods. Because the framework makes no domain-specific assumptions, it applies well beyond generative model evaluation to the broad range of scenarios where noisy pairwise comparisons must be transformed into reliable global rankings, from crowdsourced studies to expert panels. Overall, our work contributes a general foundation for more cost-efficient, interpretable, and reproducible human studies, both for evaluating AI-generated content and for preference aggregation more widely.

2 RELATED WORK

Human studies across many disciplines require ranking items. Presenting participants with a choice between two items rather than a single item with a score increases sensitivity to subtle differences and reduces variability in responses (Zerman et al., 2018; Wang et al., 2023), a benefit that holds broadly regardless of the items being judged. Consequently, leaderboards for machine learning models, along with evaluation efforts in numerous other fields, increasingly rely on pairwise comparisons. In such setups, participants indicate their preference between two alternatives, and these preferences are then aggregated to produce overall rankings. Notable examples include the Chatbot Arena (Chiang et al., 2024) and the CLIC image compression challenge (CLIC, 2025), which

¹The IHQ dataset will be released upon acceptance.

use pairwise comparisons and combine them using variants of the Bradley-Terry (BT) model, and similar pipelines appear wherever human preference must be aggregated at scale.

The Bradley-Terry model (Bradley & Terry, 1952) was originally developed to rank competitors in sports, but its scope has since grown far beyond that setting. It provides a general probabilistic framework to estimate the likelihood that one item is preferred over another. By converting pairwise comparisons into a ranking, it has become a cornerstone in studies spanning games, consumer preferences, and a wide array of other applications. Several extensions have been proposed to broaden its applicability still further. For instance, the Plackett-Luce model generalizes the BT framework from pairwise comparisons to rankings over multiple items (Plackett, 1975; Luce et al., 1959), defining a probability distribution over permutations by multiplying successive BT probabilities (Luce et al., 1959). Other modifications extend the model to additional contexts, such as home-field advantages (Agresti, 2010), ties (Rao & Kupper, 1967), or comparisons between groups rather than individuals (Huang et al., 2006).

Maximum likelihood estimation (MLE) is the standard route to inferring the parameters of the basic BT model. Zermelo (1929) introduced an iterative approach to compute these estimates that has since been widely adopted. Later, Lange et al. (2000) demonstrated that this procedure is a particular case of minorization-maximization (MM) algorithms, which iteratively optimize surrogate functions to reach a local maximum of the likelihood. Hunter (2004) extended MM algorithms to generalized BT models and established conditions guaranteeing convergence to the MLE. For the classical BT model, the optimization is convex, ensuring convergence to the global optimum (Hunter, 2004). Once the model is generalized to richer settings such as home-field advantage, multiple comparisons, or ties, however, MM algorithms may converge only to a local maximum (Hunter, 2004).

Bayesian formulations of the BT model have been explored to incorporate prior knowledge and regularization (Adams, 2005; Guiver & Snelson, 2009; Caron & Doucet, 2012). Caron & Doucet (2012) showed that MM algorithms can be interpreted as Expectation-Maximization (EM) procedures, enabling Bayesian inference via Gibbs sampling. This formulation provides tractable complete-data likelihoods and ensures convergence of the resulting Markov Chain Monte Carlo methods. More generally, the Bayesian perspective smooths posterior estimates, reduces susceptibility to local maxima, and supports principled uncertainty quantification for the estimated item skills.

Annotator quality models represent a critical extension that relaxes the assumption that all comparisons are equally reliable, an assumption that rarely holds in any human study. In crowdsourced evaluations especially, participant quality can vary widely, motivating approaches that explicitly account for annotator reliability (Chen et al., 2013). Chen et al. (2013) proposed a Bayesian model that jointly estimates both item quality and rater reliability using annotator-specific parameters. Their approach places Gaussian priors on item and rater parameters, and incorporates scaling factors in the likelihood to modulate individual annotator influence. They perform posterior inference using gradient-based optimization, which can be challenging in high-dimensional spaces due to potential local optima. Moreover, gradient-based methods offer no convergence guarantees. In contrast, our proposed EM-based method provides guaranteed monotonic likelihood improvement, independent of the domain in which it is applied. The Elo-based rater model developed by Google Research implements a simplified version of this idea. It is widely used in the Challenge on Learned Image Compression (CLIC) (CLIC, 2025) and in related research (Mentzer et al., 2020; Ballé et al., 2025).

To the best of our knowledge, no prior work combines the Bradley-Terry model with a Bayesian formulation that explicitly models rater quality while also providing closed-form EM updates. In contrast to Chen et al. (2013), who use Gaussian priors and gradient-based optimization without convergence guarantees, our approach leverages conjugate priors and an EM algorithm that ensures monotonic likelihood improvement in general. Compared to the Elo-based rater model widely used in CLIC (CLIC, 2025), which is a simplified heuristic relying on Elo-style updates, our method provides a principled and broadly applicable Bayesian treatment with uncertainty estimates and interpretable rater-quality parameters.

3 METHODOLOGY

3.1 BRADLEY-TERRY MODEL WITH RATER QUALITY

Converting a set of noisy pairwise comparisons into a reliable ranking of items is a fundamental problem that recurs throughout machine learning, statistics, and the empirical sciences. In its most general form, consider a set of K items that are repeatedly compared with one another in pairs by a set of R raters. The data, which we denote as D , consists of the outcomes of these comparisons, whatever the items happen to represent. For two items i and j of this set, Bradley & Terry (1952) suggested the following model:

$$P(i \text{ beats } j) = \frac{\lambda_i}{\lambda_i + \lambda_j} \quad (1)$$

where $\lambda_k > 0$ is a parameter associated with item $k \in \{1, 2, \dots, K\}$ that represents its skill rating. This model provides a clear and interpretable way to infer item scores from a set of observed wins and losses. However, it operates under the simplifying assumption that all comparisons are equally reliable. In any human evaluation, regardless of domain, this assumption is often violated. Participants may exhibit varying levels of expertise, attention, or diligence, leading to unreliable and inconsistent judgments.

To account for varying rater reliability in a way that does not depend on the nature of the items, we introduce a rater-specific quality parameter $q_r \in [0, 1]$. Intuitively, with probability q_r , rater r makes an informed judgment following the Bradley-Terry model. With probability $1 - q_r$, the rater guesses randomly, as if flipping a fair coin between the two items. This leads to the following mixture model for the probability that rater r ranks item i above item j :

$$P(r \text{ ranks } i \text{ above } j) = q_r \left(\frac{\lambda_i}{\lambda_i + \lambda_j} \right) + (1 - q_r) \left(\frac{1}{2} \right) \quad (2)$$

The log-likelihood of the data is given by:

$$\begin{aligned} \log P(D \mid \lambda, q) = & \sum_{r=1}^R \sum_{i=1}^K \sum_{\substack{j=1 \\ j \neq i}}^K \left[w_{r,ij} \log \left(q_r \frac{\lambda_i}{\lambda_i + \lambda_j} + (1 - q_r) \frac{1}{2} \right) \right. \\ & \left. + (n_{r,ij} - w_{r,ij}) \log \left(q_r \frac{\lambda_j}{\lambda_i + \lambda_j} + (1 - q_r) \frac{1}{2} \right) \right], \end{aligned} \quad (3)$$

where $n_{r,ij}$ denotes the total number of comparisons between items i and j by rater r , and $w_{r,ij}$ is the number of times rater r ranks item i above item j . There is no closed-form maximum likelihood estimator for this likelihood function, so the optimal parameters λ and q cannot be derived analytically. While an iterative optimization method like gradient descent could be used, it offers no guarantee of convergence. The Expectation–Maximization (EM) algorithm, by contrast, avoids learning-rate tuning and guarantees a monotonic increase of the observed-data likelihood together with convergence to a stationary point (Dempster et al., 1977), properties that hold uniformly across problem instances.

3.2 THURSTONIAN INTERPRETATION

Following Caron & Doucet (2012), we interpret the Bradley-Terry model under a Thurstonian framework (Diaconis, 1988), a general latent-variable view that applies to any pairwise comparison. In this perspective, a comparison between items i and j is conceptualized as a race, where each item has a random arrival time, Y_i and Y_j , respectively. These arrival times are assumed to follow exponential distributions:

$$Y_i \sim \mathcal{E}(\lambda_i), \quad Y_j \sim \mathcal{E}(\lambda_j), \quad (4)$$

and the item with the smaller arrival time is declared the winner. This leads directly to the standard Bradley-Terry probability:

$$P(i \text{ beats } j) = P(Y_i < Y_j) = \frac{\lambda_i}{\lambda_i + \lambda_j}. \quad (5)$$

For the EM algorithm, we introduce latent variables to simplify the complete-data likelihood. First, we define an indicator variable

$$A_{r,ij}^{(c)} \sim \text{Bernoulli}(q_r), \quad (6)$$

which denotes whether the c -th comparison of items i and j by rater r follows the Bradley-Terry model. Using these indicators, we define the latent variable $Z_{r,ij}$ as the sum of the minimal arrival times across the $n_{r,ij}$ comparisons by rater r :

$$Z_{r,ij} = \sum_{c=1}^{n_{r,ij}} A_{r,ij}^{(c)} \min(Y_i^{(c)}, Y_j^{(c)}). \quad (7)$$

Conditioned on $m_{r,ij} = \sum_{c=1}^{n_{r,ij}} A_{r,ij}^{(c)}$, the variable $Z_{r,ij}$ follows a Gamma distribution,

$$Z_{r,ij} \mid m_{r,ij} \sim \Gamma(m_{r,ij}, \lambda_i + \lambda_j), \quad (8)$$

where $\Gamma(\alpha, \beta)$ denotes the Gamma distribution with shape parameter α and inverse scale β . This Gamma-distributed latent variable formulation yields a tractable EM update while accounting for rater-specific quality, and it does so without any assumptions tied to a particular type of comparison data.

3.3 EXPECTATION-MAXIMIZATION UPDATES

The EM algorithm is an iterative method for finding maximum a posteriori (MAP) estimates for our model parameters, λ and q , by treating the rater's quality and the unobserved arrival times from the Thurstonian interpretation as latent variables. Regardless of the dataset, the algorithm is guaranteed to converge to a stationary point of the posterior distribution.

First, we specify prior distributions for the parameters. The item skills λ_k are assigned a Gamma prior, $\lambda_k \sim \Gamma(a, b)$, which is a conjugate prior for the exponential distribution. Each rater's quality parameter, q_r , is given a Beta prior, $q_r \sim B(\alpha, \beta)$. These conjugate choices keep the resulting updates in closed form for any problem size.

The core of the EM algorithm is the iterative maximization of the expected complete-data log-posterior, conditioned on the current parameter estimates (λ^*, q^*). The algorithm proceeds by alternating between two steps until convergence:

E-step: Expectation In the E-step, we compute the expected complete-data log-posterior, a function we denote as Q . This function represents the expected value of the log-posterior of all observed and latent variables, given our observed data and the current parameter estimates from the previous iteration. Independent of the specific comparison data, it is defined as:

$$Q(\lambda, q \mid \lambda^*, q^*) = \mathbb{E}_{A, Z \mid D, \lambda^*, q^*} [\ell_c(\lambda, q; D, Z, A) + \log P(\lambda) + \log P(q)]. \quad (9)$$

The complete-data log-likelihood, ℓ_c , is further broken down into three components.

$$\ell_c(\lambda, q; D, Z, A) = \log P(Z \mid D, A, \lambda, q) + \log P(A \mid D, \lambda, q) + \log P(D \mid \lambda, q). \quad (10)$$

M-step: Maximization This step updates the model parameters by maximizing the Q function. By leveraging the expected values from the E-step, the M-step transforms the original, generally intractable optimization problem into simpler, closed-form updates that apply uniformly to any dataset. The key quantity used in both updates is the posterior probability that a given comparison from rater r follows the Bradley-Terry model. This quantity, denoted as γ , represents the weight of a rater's judgment based on how much it aligns with the model's current predictions. It is given by:

$$\gamma_{r,ij}^{(t-1)} = \frac{q_r^{(t-1)} y_{ij}^{(t-1)}}{q_r^{(t-1)} y_{ij}^{(t-1)} + (1 - q_r^{(t-1)}) \frac{1}{2}}, \quad (11)$$

where $y_{ij}^{(t-1)} = \frac{\lambda_i^{(t-1)}}{\lambda_i^{(t-1)} + \lambda_j^{(t-1)}}$ is the Bradley-Terry probability that item i beats item j . This posterior probability γ captures our confidence that a comparison was meaningful rather than random, given

the current parameter estimates, and this interpretation carries over to any application of the model. Higher γ values indicate more trustworthy comparisons.

The new estimate for a rater’s quality, q_r , is a weighted average. The numerator sums the “effective number of wins” for that rater, where each win is weighted by the posterior probability (γ) that it reflected a meaningful, non-random judgment, combined with the hyperparameters of the Beta prior to regularize the estimate. The denominator normalizes this sum by the total number of comparisons and the prior parameters. Intuitively, and for any type of rater, this update raises a rater’s quality score when their judgments frequently align with the model’s predictions. The update is given by:

$$q_r^{(t)} = \frac{\sum_{i=1}^K \sum_{j=i+1}^K \left[w_{r,ij} \gamma_{r,ij}^{(t-1)} + w_{r,ji} \gamma_{r,ji}^{(t-1)} \right] + (\alpha - 1)}{n_r + \alpha + \beta - 2}, \quad (12)$$

where n_r is the total number of comparisons by rater r .

The new estimate for an item’s skill, λ_i , is a ratio that balances two key quantities. The numerator sums the “effective wins” for item i across all raters, where each win is again weighted by the rater’s quality (γ); this term represents the total positive evidence for item i . The denominator accounts for all comparisons item i was involved in and acts as a normalizing factor. Both terms are regularized by the Gamma prior hyperparameters, and the form of the update is the same whatever the items represent. The update is given by:

$$\lambda_i^{(t)} = \frac{\sum_{r=1}^R \left[\sum_{j=1, j \neq i}^K w_{r,ij} \gamma_{r,ij}^{(t-1)} \right] + (a - 1)}{\sum_{j=1, j \neq i}^K \left[\frac{\sum_{r=1}^R \left[w_{r,ij} \gamma_{r,ij}^{(t-1)} + w_{r,ji} \gamma_{r,ji}^{(t-1)} \right]}{\lambda_i^{(t-1)} + \lambda_j^{(t-1)}} \right] + b}. \quad (13)$$

The derivation of the Expectation Maximization algorithm is provided in Appendix B.

4 EXPERIMENTAL RESULTS

To evaluate the performance of our **Bayesian Bradley-Terry** model with rater **Quality (BBQ)**, we conduct a broad series of experiments comparing it against two baselines: Bayesian Bradley-Terry (Bayes-BT) (Caron & Doucet, 2012) and a gradient descent-based BT model that incorporates rater quality (Crowd-BT) (Chen et al., 2013). For Crowd-BT, we use the implementation provided by Google Research (Google, 2025), which was employed both by CLIC (2025) and Ballé et al. (2025). The hyperparameters used in our experiments can be found in Section C.

To probe how broadly the conclusions hold, we evaluate several Bradley-Terry (BT) variants across a diverse set of datasets that span two very different modalities: human preference benchmarks for large language models (HUMAINE (Team, 2025), MT-Bench (Zheng et al., 2023)) and image compression (WD (Ballé et al., 2025), HiFiC (Mentzer et al., 2020), ConHa (Aczel & Wattenhofer, 2024)), together with a newly collected inhomogeneous rater quality (IHQ) dataset from the CLIC2024 image test set (CLIC, 2025). To study the effect of rater quality, we partition the IHQ dataset into two subsets, screened and unscreened. This setup lets us analyze how low-quality, high-quality, and mixed comparisons affect model performance across the full range of conditions encountered in practice. Details of the IHQ dataset creation are given in Section E, and a summary of all datasets, including comparisons, raters, and items, is provided in Section D and Table 2.

Ground truth rankings are not available for real-world datasets of any kind. We therefore first validated the models on simulated data, where all methods recover the same ordering as the number of samples increases. For crowdsourced datasets, we approximate the ground truth by the ranking obtained on the whole dataset. We check that this is a reasonable proxy for ordering across our real-world datasets by examining the top-1 item in each ranking. For every dataset except IHQ-unscreened, all three models recover the same top-ranked item. On the IHQ-unscreened dataset, Crowd-BT and Bayes-BT fail to identify the reference image as the highest-quality item, whereas BBQ succeeds.

A reliable aggregation method should reproduce the same ordering if the study is repeated, a requirement that applies to any evaluation. We approximate this stability using bootstrapping, which

estimates the variability in the rankings. Details on the bootstrapping procedure are given in Section F. Note that the WD and HiFiC studies employed active selection of comparison pairs: HiFiC used a binary search strategy, while WD selected pairs based on maximum information gain. For this reason, the bootstrapping results on these two datasets should be interpreted with some caution.

We report two complementary metrics that together capture both ends of the ranking. Top-1 agreement is the fraction of bootstrapped samples that identify the same best item as the full dataset. Kendall’s Tau (τ) measures ordinal correlation between rankings (Kendall, 1938), with higher values indicating more stable results. Top-1 agreement is most relevant when only the best item matters, while Kendall’s Tau assesses the stability of the ranking as a whole.

4.1 PERFORMANCE ACROSS DATASETS

We compare the models across all datasets to assess ranking accuracy and stability under widely varying conditions. Top-1 agreement and Kendall’s Tau are summarized in Table 1. As a general trend, datasets with more comparisons per model, such as MT-Bench, WD, and HiFiC, are easier. HUMAINE is an instructive exception, underscoring that factors beyond the sheer number of comparisons, such as rater consistency and diversity, can shape model performance in any given study.

Taken together, the results show that BBQ delivers superior stability and robustness across this diverse collection of datasets. It identifies the top-performing item most frequently, being the shared best on three datasets. It also recovers the overall ranking most accurately on five of the eight datasets and ranks second on the remaining three, indicating consistent behavior across both language and vision settings.

The three datasets where BBQ ranks second in Kendall’s Tau all share a common characteristic: high-quality, homogeneous rater sets. On MT-Bench, all models perfectly recover the top item. The WD dataset, collected by Ballé et al. (2025), includes only five raters, likely the paper’s authors, suggesting exceptionally careful evaluation. In IHQ-screened, raters were explicitly filtered for quality. Whenever rater quality is uniformly high, explicitly modeling rater reliability offers little to gain, and the advantage of BBQ over simpler models naturally narrows.

The IHQ experiments, which span both screened and unscreened rater subsets, reveal a clear and generalizable pattern. BBQ substantially outperforms Crowd-BT and Bayes-BT whenever low-quality raters are present, as in CLIC-all and CLIC-unscreened. All three models attain their best Top-1 accuracy on the screened subset, underscoring the value of rater selection. Although Crowd-BT explicitly models rater quality, its performance drops noticeably on the full dataset, likely because crowdsourced raters each provide fewer than 40 comparisons, which increases susceptibility to noisy annotations. BBQ, by contrast, retains strong performance even without screening, with only a minor decrease in Top-1 accuracy, demonstrating a robustness to low-quality raters that should carry over to other noisy, large-scale settings.

Nonetheless, achieving uniformly high rater quality is difficult in essentially any large study. Large-scale crowdsourcing introduces variability, screening procedures are costly, and subjective factors can affect even diligent annotators. In all such settings, BBQ offers a principled way to make use of partially reliable raters while reducing the impact of noisy contributions.

4.2 SCALING WITH RATERS AND COMPARISONS

As observed in Table 1, datasets with more comparisons per model generally yield better performance, a relationship that holds broadly. The total number of comparisons can be increased in two ways: by adding more raters, or by having each rater perform more comparisons.

Figure 1 disentangles the impact of these two factors on the three BT variants. The left column shows performance as a function of the number of raters (with the number of comparisons per rater fixed at the maximum), while the right column shows performance as a function of the number of comparisons per rater (with the number of raters fixed at the maximum). Top-1 accuracy separates the models clearly, whereas Kendall’s τ remains similar across them. Crowd-BT fails to converge with only one or two raters, highlighting the general advantage of the EM algorithm, which provides convergence guarantees that gradient descent lacks.

Table 1: Performance of different BT-based aggregation methods across a diverse set of language and vision datasets. Top: Top-1 agreement [%], Bottom: Kendall’s Tau. Across these varied conditions, BBQ most frequently identifies the top-performing item, recovers the best overall ranking on more than half of the datasets, and ranks second on the rest.

	HUMAINE	MT-Bench	WD	HiFiC	ConHa	IHQ		
						all	scr.	unscr.
Top-1 Agreement [%]								
Crowd-BT	85.60	100.00	99.29	98.63	<u>66.59</u>	<u>85.31</u>	98.57	<u>33.15</u>
Bayes-BT	<u>97.30</u>	100.00	100.00	100.00	57.30	75.07	<u>98.90</u>	24.32
BBQ (ours)	97.50	100.00	100.00	100.00	77.12	99.32	99.80	61.92
Kendall’s Tau								
Crowd-BT	0.9385	0.9743	0.9279	<u>0.9366</u>	0.9180	0.9245	0.9171	0.8482
Bayes-BT	<u>0.9443</u>	0.9569	0.9459	0.9293	0.9110	0.9240	0.9116	<u>0.8507</u>
BBQ (ours)	0.9463	<u>0.9675</u>	<u>0.9359</u>	0.9525	0.9265	0.9270	<u>0.9132</u>	0.8563

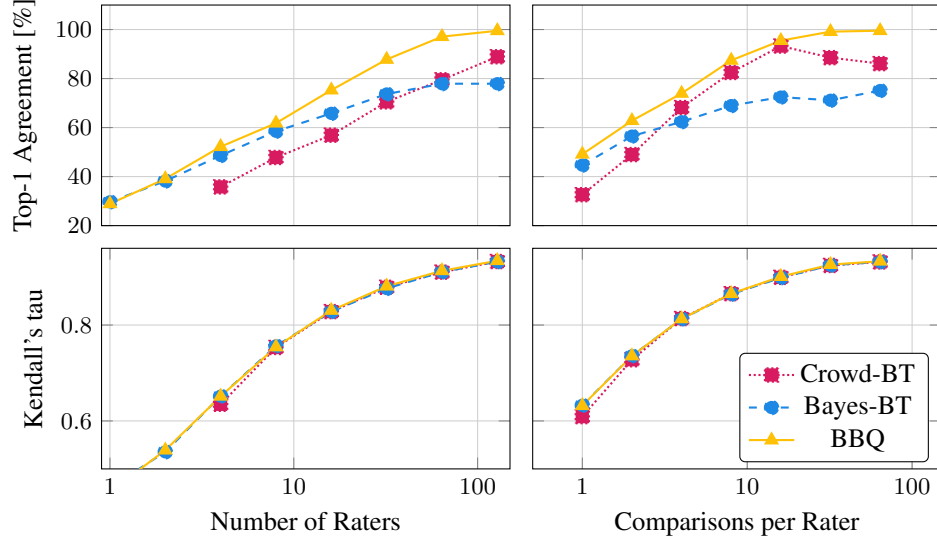


Figure 1: Scaling behavior of Bradley–Terry variants (Crowd-BT (Caron & Doucet, 2012), Bayes-BT (Chen et al., 2013), BBQ (ours)) on the IHQ-all dataset, illustrating how the methods behave across the full range from data-scarce to data-rich regimes. **Left:** Performance vs. number of raters. **Right:** Performance vs. number of comparisons per rater. Both Top-1 agreement and Kendall’s τ improve markedly as either more raters or more comparisons become available. Top-1 agreement clearly separates the models, whereas Kendall’s τ stays similar across them. Crowd-BT fails to converge with very few raters, highlighting the EM algorithm’s advantage. Crowd-BT and BBQ behave similarly under sparse data, but BBQ overtakes Bayes-BT as the number of raters or comparisons grows.

Crowd-BT and BBQ perform similarly when either the number of raters or the number of comparisons per rater is small. BBQ needs both multiple raters and multiple comparisons per rater before it can effectively distinguish between rater qualities. As either quantity grows, BBQ increasingly outperforms Bayes-BT. Bayes-BT tends to trail overall but can surpass Crowd-BT in the sparse regime of limited data per rater or few raters, precisely where Crowd-BT cannot reliably estimate rater quality.

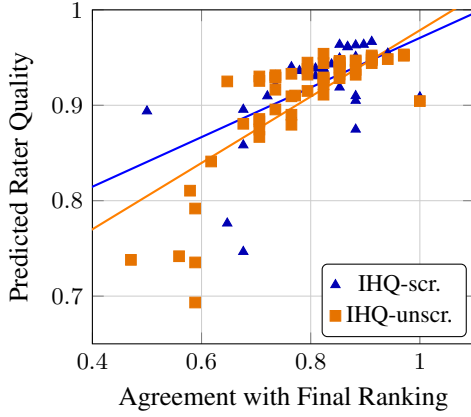


Figure 2: Scatter plot of rater agreement with the final ranking (x-axis) against the predicted rater quality (y-axis) for the IHQ datasets, showing that the model recovers meaningful per-rater reliability. Each point corresponds to an individual rater. Triangles denote the filtered dataset and squares the unfiltered dataset.

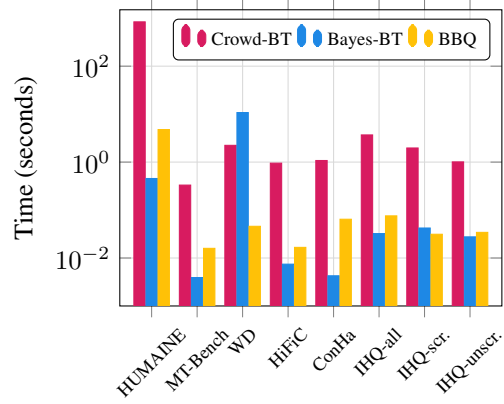


Figure 3: Average computation time in seconds (log scale) for the three models, measured on a single bootstrapped sample across all eight datasets. While Crowd-BT can require substantial computation time on some datasets, BBQ remains fast across the entire range of datasets.

4.3 RATER QUALITY

Figure 2 relates agreement with the final ranking to predicted rater quality. We observe a clear positive correlation: raters who agree more closely with the consensus ranking are assigned higher quality by the model, indicating that the learned quality parameters are meaningful rather than artifacts of a particular subset. For the filtered dataset, the Pearson correlation is $r = 0.551$ across 50 raters. For the unfiltered dataset it is stronger, at $r = 0.724$ across 62 raters.

As expected, the unscreened dataset contains several raters with both lower agreement and lower predicted quality. Some raters in the IHQ-unscreened dataset even fall below the level of random guessing (50% agreement), systematically disagreeing with the majority. This illustrates, in a concrete case, the broader importance of modeling rater quality when aggregating any pairwise comparison data. BBQ successfully identifies such raters and assigns them lower quality scores, reducing their influence on the final ranking and mitigating the noise they introduce.

4.4 COMPUTATIONAL EFFICIENCY

Figure 3 reports the average computation time in seconds each method needs to process a single bootstrapped sample across the different datasets. These timings offer a practical perspective on the feasibility of the methods in real-world evaluation scenarios of varying scale. BBQ converges within a few seconds on every dataset. Crowd-BT is slower than BBQ across the board, with particularly long runtimes on datasets that have many comparisons, such as HUMAINE, where convergence takes around 15 minutes. Bayes-BT converges slowly on the WD dataset, though on the others it is slightly faster than BBQ. The efficiency of BBQ stems from the closed-form EM updates derived in our method, which enable rapid convergence even as datasets grow large.

It is also worth noting that Crowd-BT was highly optimized and implemented in C (Kernighan & Ritchie, 1988), whereas BBQ was implemented in plain NumPy without any special optimizations. This suggests that BBQ could be made even faster with a compiled or vectorized implementation. Consequently, BBQ is not only more robust and stable but also highly practical for large-scale human preference studies and, more generally, for any application that requires repeated bootstrapping. This combination of accuracy, stability, and speed makes BBQ a compelling choice for a wide variety of real-world deployments.

4.5 CONFIDENCE INTERVALS

Beyond ranking accuracy, we assess how Crowd-BT, Bayes-BT, and BBQ estimate uncertainty in item rankings, a capability that matters whenever conclusions must be tested for significance. In a controlled simulation, raters provide pairwise comparisons of equally skilled items (H_0). Each model estimates skills and constructs uncertainty intervals to test for significant differences (Section G). The resulting Type I error rates are shown in Figure 6. At the 99% confidence level, the expected error rate is 1%. Crowd-BT and BBQ align closely with this target, yielding slightly lower rates, whereas Bayes-BT is more conservative at $\sim 0.1\%$. BBQ therefore provides well-calibrated uncertainty estimates, on par with Crowd-BT, while Bayes-BT underestimates false positives.

5 LIMITATIONS

Although BBQ effectively models rater quality, its advantages diminish in the special case where all raters are uniformly reliable. In such homogeneous datasets, modeling variability adds little benefit. Ensuring consistently high-quality raters, however, typically demands substantial cost and effort that is often infeasible at scale. This tension indicates that BBQ is most valuable in the realistic, noisy conditions that dominate large-scale evaluation, and comparatively less so in the narrower case of carefully curated studies.

A second consideration concerns the form of feedback the framework currently ingests. BBQ operates on pairwise comparisons, whereas some human evaluation studies instead collect ratings, full rankings, or multi-way inputs. Extending the model to accommodate these additional forms of feedback is a natural direction that would widen its already broad applicability.

From a methodological perspective, our experiments rely on a finite number of bootstrapped comparisons. While this offers a practical measure of stability, still larger studies would help further validate robustness across the full spectrum of deployment conditions. The model also assumes independence across comparisons and does not model contextual or order effects, which can influence human judgments in some studies.

Finally, although BBQ scales well computationally, very large numbers of items or raters could still pose challenges in the absence of an optimized or compiled implementation.

6 CONCLUSION

We introduced BBQ, a general Bayesian Bradley-Terry model that jointly estimates item quality and rater reliability. Our core contribution is the derivation of an Expectation-Maximization (EM) algorithm that simultaneously estimates item skills and individual rater quality, effectively down-weighting or removing the influence of unreliable participants. By explicitly modeling rater quality, the method produces more stable and accurate rankings across settings. The EM algorithm guarantees rapid convergence and monotonic likelihood improvement, addressing the limitations of gradient-based approaches regardless of the application.

Across a diverse collection of language and vision datasets, BBQ consistently achieved high Top-1 agreement and superior Kendall’s Tau, demonstrating robustness and reliability across modalities. The model excels in scenarios with noisy or crowdsourced raters, maintaining accuracy even when raters contribute few comparisons. It is particularly effective in large-scale settings where many raters are non-experts or vary widely in attentiveness and diligence, conditions that are common far beyond the studies examined here. Our results underline that incorporating rater quality becomes especially crucial whenever evaluation quality is heterogeneous or partially unreliable. The Bayesian framework further supplies principled uncertainty estimates for item scores, enabling interpretable comparisons across studies and settings. We also show that the model’s error bars are well-calibrated and can be used to assess whether differences between items are statistically significant. Predicted rater quality aligns with agreement to the final rankings, validating the model’s ability to identify reliable evaluators in general.

Overall, BBQ advances human evaluation methodology by offering a practical, interpretable, and broadly generalizable approach to aggregating noisy pairwise comparisons. This work marks a significant step toward more cost-effective, interpretable, and reproducible human studies, both for

540 evaluating AI-generated content and for the many other domains in which pairwise preferences must
541 be turned into reliable rankings.
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A USAGE OF LLMs

During the preparation of this paper, we utilized large language models (LLMs) as supportive tools. ChatGPT, Claude, Gemini, and Grammarly assisted with spellchecking, refining phrasing, and condensing text to enhance clarity and readability. Additionally, ChatGPT, Claude, and Cursor were used for code analysis, completion, and the generation of visualizations to aid our development workflow. These tools were employed solely as auxiliary aids, while all core research ideas, experimental design, and interpretation of results were developed independently by the authors.

B BAYESIAN BT WITH RATER QUALITY DERIVATION

We consider the general setting of R raters comparing K items. Each rater r has quality q_r , meaning that with probability q_r they follow the Bradley–Terry model, and with probability $1 - q_r$ they choose randomly. The following is a standard EM (Dempster et al., 1977) derivation that incorporates these rater-specific reliabilities into the Bayesian estimation framework, and it applies to any such comparison problem.

B.1 MODEL DEFINITION

Notation:

- $\lambda = (\lambda_1, \dots, \lambda_K)$: skill parameters for the K items.
- $q = (q_1, \dots, q_R)$: quality parameters for the R raters.
- $Y_i \sim \mathcal{E}(\lambda_i)$: latent arrival time associated with item i .
- $A_{r,ij}^{(c)} \sim \text{Bernoulli}(q_r)$: indicator that the c -th comparison of pair (i, j) by rater r follows the Bradley-Terry model.
- $n_{r,ij}$: total number of comparisons of (i, j) by rater r .
- $n_{ij} = \sum_{r=1}^R n_{r,ij}$: total number of comparisons of (i, j) .
- $w_{r,ij}$: number of times rater r ranked i above j .
- $w_{ij} = \sum_{r=1}^R w_{r,ij}$: total number of times i was ranked above j .
- $m_{r,ij} = \sum_{c=1}^{n_{r,ij}} A_{r,ij}^{(c)}$: number of Bradley-Terry-model comparisons of (i, j) by rater r .
- $m_{ij} = \sum_{r=1}^R m_{r,ij}$: total number of Bradley-Terry-model comparisons of (i, j) .
- $v_{r,ij} = \sum_{c=1}^{w_{r,ij}} A_{r,ij}^{(c)}$: number of Bradley-Terry-model wins of i over j by rater r .
- $v_{ij} = \sum_{r=1}^R v_{r,ij}$: total number of Bradley-Terry-model wins of i over j .
- $Z_{r,ij} = \sum_{c=1}^{n_{r,ij}} A_{r,ij}^{(c)} \min(Y_i^{(c)}, Y_j^{(c)})$: sum of minimal arrival times, with

$$Z_{r,ij} \mid m_{r,ij} \sim \Gamma(m_{r,ij}, \lambda_i + \lambda_j).$$

Bradley-Terry probability:

$$P(i \text{ beats } j) = \frac{\lambda_i}{\lambda_i + \lambda_j}.$$

Mixture with rater quality:

$$P(r \text{ ranks } i \text{ above } j) = q_r \frac{\lambda_i}{\lambda_i + \lambda_j} + (1 - q_r) \frac{1}{2}.$$

Latent exponential view:

$$Y_i \sim \mathcal{E}(\lambda_i), \quad Y_j \sim \mathcal{E}(\lambda_j), \quad P(i \text{ beats } j) = P(Y_i < Y_j).$$

B.2 COMPLETE-DATA LOG-LIKELIHOOD

We need to compute:

$$\begin{aligned}\ell_c(\lambda, q; D, Z, A) &= \log P(D, Z, A \mid \lambda, q) \\ &= \log P(Z \mid D, A, \lambda, q) + \log P(A \mid D, \lambda, q) + \log P(D \mid \lambda, q)\end{aligned}$$

Log-Likelihood of $P(Z \mid A, \lambda)$:

$$\begin{aligned}\log P(Z \mid D, A, \lambda, q) &= \sum_{i=1}^K \sum_{j=i+1}^K \left[m_{ij} \log(\lambda_i + \lambda_j) - (\lambda_i + \lambda_j) z_{ij} \right. \\ &\quad \left. + (m_{ij} - 1) \log z_{ij} - \log \Gamma(m_{ij}) \right]\end{aligned}$$

Log-Likelihood of $P(A \mid q)$:

$$\log P(A \mid D, \lambda, q) = \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K \left[m_{r,ij} \log q_r + (n_{r,ij} - m_{r,ij}) \log(1 - q_r) \right]$$

Log-Likelihood of $P(D \mid \lambda, A)$:

$$\begin{aligned}\log P(D \mid \lambda, q) &= \sum_{r=1}^R \sum_{i=1}^K \sum_{\substack{j=1 \\ j \neq i}}^K \left[v_{r,ij} \log \frac{\lambda_i}{\lambda_i + \lambda_j} + (w_{r,ij} - v_{r,ij}) \log \frac{1}{2} \right] \\ &= \sum_{r=1}^R \sum_{i=1}^K [v_{ir} \log \lambda_i] \\ &\quad - \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K [(v_{r,ij} + v_{r,ji}) \log(\lambda_i + \lambda_j) + (w_{r,ij} + w_{r,ji} - v_{r,ij} - v_{r,ji}) \log 2] \\ &= \sum_{i=1}^K [v_i \log \lambda_i] \\ &\quad - \sum_{i=1}^K \sum_{j=i+1}^K [m_{ij} \log(\lambda_i + \lambda_j) + (n_{ij} - m_{ij}) \log 2]\end{aligned}$$

Complete-Data Log-Likelihood:

$$\begin{aligned}
\ell_c(\lambda, q; D, Z, A) &= \log P(Z \mid D, A, \lambda, q) + \log P(A \mid D, \lambda, q) + \log P(D \mid \lambda, q) \\
&= \sum_{i=1}^K \sum_{j=i+1}^K \left[m_{ij} \log(\lambda_i + \lambda_j) - (\lambda_i + \lambda_j) z_{ij} + (m_{ij} - 1) \log z_{ij} - \log \Gamma(m_{ij}) \right] \\
&\quad + \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K \left[m_{r,ij} \log q_r + (n_{r,ij} - m_{r,ij}) \log(1 - q_r) \right] \\
&\quad + \sum_{i=1}^K [v_i \log \lambda_i] \\
&\quad - \sum_{i=1}^K \sum_{j=i+1}^K [m_{ij} \log(\lambda_i + \lambda_j) + (n_{ij} - m_{ij}) \log 2] \\
&= \sum_{i=1}^K \sum_{j=i+1}^K \left[(n_{ij} - m_{ij}) \log 2 - (\lambda_i + \lambda_j) z_{ij} + (m_{ij} - 1) \log z_{ij} - \log \Gamma(m_{ij}) \right] \\
&\quad + \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K \left[m_{r,ij} \log q_r + (n_{r,ij} - m_{r,ij}) \log(1 - q_r) \right] \\
&\quad + \sum_{i=1}^K [v_i \log \lambda_i]
\end{aligned}$$

B.3 EXPECTATION STEP

We introduce conjugate priors: Gamma distribution $\lambda_i \sim \Gamma(a, b)$ for each item i , and Beta $q_r \sim B(\alpha, \beta)$ for each rater r .

The Q -function is the expectation of the complete-data log-posterior:

$$Q(\lambda, q \mid \lambda^*, q^*) = \mathbb{E}_{A, Z \mid D, \lambda^*, q^*} [\ell_c(\lambda, q; D, Z, A) + \log P(\lambda) + \log P(q)].$$

Posterior probability of a Bradley-Terry-consistent annotation:

$$\begin{aligned}
P(A_{r,ij}^{(k)} = 1 \mid i \succ j) &= \frac{P(i \succ j \mid A = 1) P(A = 1)}{P(i \succ j \mid A = 1) P(A = 1) + P(i \succ j \mid A = 0) P(A = 0)} \\
&= \frac{\left(\frac{\lambda_i^*}{\lambda_i^* + \lambda_j^*} \right) q_r^*}{\left(\frac{\lambda_i^*}{\lambda_i^* + \lambda_j^*} \right) q_r^* + \frac{1}{2} (1 - q_r^*)} = \gamma_{r,ij}^*.
\end{aligned}$$

Expected sufficient statistics:

$$\begin{aligned}
\mathbb{E}[m_{r,ij}] &= w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^* \\
\mathbb{E}[m_{ij}] &= \sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) \\
\mathbb{E}[v_{r,ij}] &= w_{r,ij} \gamma_{r,ij}^* \\
\mathbb{E}[v_{ir}] &= \sum_{j: j \neq i}^K w_{r,ij} \gamma_{r,ij}^* \\
\mathbb{E}[v_i] &= \sum_{r=1}^R \sum_{j: j \neq i}^K w_{r,ij} \gamma_{r,ij}^* \\
\mathbb{E}[z_{ij}] &= \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{\lambda_i^* + \lambda_j^*}.
\end{aligned}$$

Expected complete-data log-posterior:

$$\begin{aligned}\mathbb{E}[\ell_c] = & \sum_{i=1}^K \sum_{j=i+1}^K \left[-(\lambda_i + \lambda_j) \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{\lambda_i^* + \lambda_j^*} \right] \\ & + \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K \left[(w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) \log q_r \right. \\ & \quad \left. + (n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)) \log(1 - q_r) \right] \\ & + \sum_{i=1}^K \sum_{\substack{j=1 \\ j \neq i}}^K w_{r,ij} \gamma_{r,ij}^* \log \lambda_i \\ & + \text{const.}\end{aligned}$$

Priors contribute:

$$\begin{aligned}\mathbb{E}[\log P(\lambda)] &= \sum_{i=1}^K [(a-1) \log \lambda_i - b \lambda_i], \\ \mathbb{E}[\log P(q)] &= \sum_{r=1}^R [(\alpha-1) \log q_r + (\beta-1) \log(1 - q_r)].\end{aligned}$$

Final Q -function:

$$\begin{aligned}Q(\lambda, q \mid \lambda^*, q^*) = & \sum_{i=1}^K \sum_{j=i+1}^K \left[-(\lambda_i + \lambda_j) \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{\lambda_i^* + \lambda_j^*} \right] \\ & + \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K \left[(w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) \log q_r \right. \\ & \quad \left. + (n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)) \log(1 - q_r) \right] \\ & + \sum_{i=1}^K \sum_{\substack{j=1 \\ j \neq i}}^K w_{r,ij} \gamma_{r,ij}^* \log \lambda_i \\ & + \sum_{i=1}^K [(a-1) \log \lambda_i - b \lambda_i] \\ & + \sum_{r=1}^R [(\alpha-1) \log q_r + (\beta-1) \log(1 - q_r)] \\ & + \text{const.}\end{aligned}$$

where

$$\gamma_{r,ij}^* = \frac{q_r^* \frac{\lambda_i^*}{\lambda_i^* + \lambda_j^*}}{q_r^* \frac{\lambda_i^*}{\lambda_i^* + \lambda_j^*} + (1 - q_r^*) \frac{1}{2}}.$$

B.4 M-STEP

The M-step maximizes the Q -function w.r.t. the parameters (λ, q) , holding the expectations computed in the E-step fixed.

Update for q_r The update for each rater quality q_r is obtained by maximizing Q with respect to q_r (including the Beta prior).

$$\begin{aligned}
Q(q \mid \lambda^*, q^*) &= \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K \left[(w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) \log q_r \right. \\
&\quad \left. + (n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)) \log(1 - q_r) \right] \\
&\quad + (\alpha - 1) \log q_r + (\beta - 1) \log(1 - q_r) + \text{const.} \\
\frac{\partial Q}{\partial q_r} &= \sum_{i=1}^K \sum_{j=i+1}^K \left[\frac{w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*}{q_r} - \frac{n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{1 - q_r} \right] \\
&\quad + \frac{\alpha - 1}{q_r} - \frac{\beta - 1}{1 - q_r} \\
&\stackrel{0}{=} \frac{\sum_{i=1}^K \sum_{j=i+1}^K (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) + (\alpha - 1)}{q_r} \\
&\quad - \frac{\sum_{i=1}^K \sum_{j=i+1}^K (n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)) + (\beta - 1)}{1 - q_r} \\
&\Rightarrow (1 - q_r) \left(\sum_{i=1}^K \sum_{j=i+1}^K (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) + (\alpha - 1) \right) \\
&\Rightarrow q_r \left(\sum_{i=1}^K \sum_{j=i+1}^K (n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)) + (\beta - 1) \right) \\
&\Rightarrow q_r = \frac{\sum_{i=1}^K \sum_{j=i+1}^K (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) + (\alpha - 1)}{\sum_{i=1}^K \sum_{j=i+1}^K (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^* + n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)) + (\beta + \alpha - 2)} \\
&\Rightarrow q_r^{(t)} = \frac{\sum_{i=1}^K \sum_{j=i+1}^K (w_{r,ij} \gamma_{r,ij}^{(t-1)} + w_{r,ji} \gamma_{r,ji}^{(t-1)}) + (\alpha - 1)}{n_r + \beta + \alpha - 2}
\end{aligned}$$

where the last line gives the explicit update at iteration t .

Update for λ_i The update for each item skill λ_i is obtained by maximizing Q with respect to λ_i (including the Gamma prior).

$$\begin{aligned}
Q(\lambda \mid \lambda^*, q^*) &= \sum_{i=1}^K \sum_{j=i+1}^K -(\lambda_i + \lambda_j) \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{\lambda_i^* + \lambda_j^*} \\
&\quad + \sum_{r=1}^R \sum_{i=1}^K \sum_{\substack{j=1 \\ j \neq i}}^K w_{r,ij} \gamma_{r,ij}^* \log \lambda_i \\
&\quad + \sum_{i=1}^K [(a - 1) \log \lambda_i - b \lambda_i] + \text{const.}
\end{aligned}$$

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$$\begin{aligned}
\frac{\partial Q}{\partial \lambda_i} &= - \sum_{j \neq i}^K \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{\lambda_i^* + \lambda_j^*} + \sum_{r=1}^R \frac{\sum_{j \neq i}^K w_{r,ij} \gamma_{r,ij}^*}{\lambda_i} + \frac{a-1}{\lambda_i} - b \\
0 &\stackrel{!}{=} \frac{\sum_{r=1}^R \sum_{j \neq i}^K w_{r,ij} \gamma_{r,ij}^* + (a-1)}{\lambda_i} - \sum_{j \neq i}^K \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{\lambda_i^* + \lambda_j^*} - b \\
\Rightarrow \lambda_i^{(t)} &= \frac{\sum_{r=1}^R \sum_{j \neq i}^K w_{r,ij} \gamma_{r,ij}^{(t-1)} + (a-1)}{\sum_{j \neq i}^K \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^{(t-1)} + w_{r,ji} \gamma_{r,ji}^{(t-1)})}{\lambda_i^{(t-1)} + \lambda_j^{(t-1)}} + b} \\
\gamma_{r,ij}^{(t-1)} &= \frac{q_r^{(t-1)} \frac{\lambda_i^{(t-1)}}{\lambda_i^{(t-1)} + \lambda_j^{(t-1)}}}{q_r^{(t-1)} \frac{\lambda_i^{(t-1)}}{\lambda_i^{(t-1)} + \lambda_j^{(t-1)}} + (1 - q_r^{(t-1)}) \frac{1}{2}}
\end{aligned}$$

where the last line gives the explicit update at iteration t .

C HYPERPARAMETERS

For Crowd-BT, we use the implementation of Google (2025) with its default hyperparameters throughout.

The only hyperparameters in the Bayes-BT and BBQ models are the prior distribution parameters and the stopping thresholds, and we keep a single setting fixed across all datasets. For both Bayes-BT and BBQ, we consider the model converged when no ELO score changes by more than 1 between two iterations. The ELO score is obtained from the skill parameter λ as:

$$\text{ELO} = \log(\text{skill}) \cdot \text{ELO_SCALE_FACTOR}. \quad (14)$$

where we set the ELO_SCALE_FACTOR to 400.

For the skill parameters in both Bayes-BT and BBQ, we use the same gamma prior with shape 5 and rate 0.1 across every dataset. For BBQ, the beta prior on the rater quality likewise uses a single fixed setting, with $\alpha = 10$ and $\beta = 2$.

D DATASETS

To test the generality of our conclusions, we evaluate the models on a diverse set of human preference datasets that together span both the language and image domains. Table 2 provides an overview.

For natural language, we use the HUMAINE dataset (Team, 2025), a large-scale benchmark with over 100k comparisons, and MT-Bench (Zheng et al., 2023), which provides model comparison judgments from crowd workers on multi-turn dialogues.

For image compression, we consider three further datasets: WD (Ballé et al., 2025), a dense expert-labeled dataset with thousands of comparisons per rater; HiFiC (Mentzer et al., 2020), which evaluates learned image codecs; and ConHa (Aczel & Wattenhofer, 2024), which focuses on conditional generative models. These datasets differ substantially in scale, number of raters, and rater expertise, providing a broad testbed for assessing robustness across regimes.

Finally, we introduce the inhomogeneous rater quality (**IHQ**) dataset, obtained from a user study on the CLIC2024 (CLIC, 2025) data conducted via the *Mabyduck* platform. It contains two-alternative-forced-choice (2AFC) judgments across 28 generative image compression models. To isolate the impact of rater quality, we provide two subsets: (i) **screened**, where raters passed pre-screening checks for attention and display quality, and (ii) **unscreened**, where all raters are included. The screened subset represents higher-quality raters, whereas the unscreened subset better reflects the noisy conditions typical of large-scale human evaluation in general.

Table 2: Summary of the datasets used in our experiments, spanning both language and image domains and a wide range of scales. The IHQ dataset was collected by us on the Mabyduck platform and includes both 2AFC and 3AFC settings. For this dataset, we provide screened and unscreened subsets: the screened subset includes only raters who passed pre-screening tests for attention and display quality, ensuring higher reliability, while the unscreened subset includes all raters.

dataset	# comparisons	# raters	# models	comp/rater	comp/model
HUMAINE (Team, 2025)	105,220	1,977	27	53.2	3897.0
MT-Bench (Zheng et al., 2023)	3,355	65	6	51.6	559.2
WD (Ballé et al., 2025)	16,659	5	30	3331.8	555.3
HiFiC (Mentzer et al., 2020)	5,220	20	9	261.0	580.0
ConHa (Aczel & Wattenhofer, 2024)	1,531	40	8	38.3	191.4
IHQ-all	4,074	112	28	36.4	145.5
IHQ-screened	2,012	50	28	40.2	71.9
IHQ-unscreened	2,062	62	28	33.3	73.6

Together, this collection lets us study the full continuum from large-scale, relatively clean benchmarks to smaller, noisier settings in which modeling rater reliability is critical.

E USER STUDY PLATFORM

All pairwise comparisons on the CLIC2024 (CLIC, 2025) dataset were collected using the Mabyduck platform (Ltd., 2025), following a protocol that is readily transferable to other comparison studies. A screenshot of the platform is shown in Figure 4. The task asks: “Which image looks more similar to the reference image?” Ties are not allowed, and all pairs were selected uniformly at random.

To study the effect of rater quality, we split the IHQ dataset into screened and unscreened subsets. Screened raters passed pre-tests for color vision, display contrast, and sensitivity to subtle differences, and completed training comparisons in which they chose the higher-quality image to reinforce the evaluation criteria; such screening protocols apply broadly to visual comparison studies.

Figure 5 shows four example images from the pre-screening process. The first two are standard color blindness tests, in which raters must correctly identify the number shown in each pattern. The final two are shape detection tests designed to evaluate the raters’ sensitivity to low-contrast objects: one features a light gray shape on a white background, and the other a dark gray shape on a black background. Such pre-screening tests help ensure that only raters with adequate visual capabilities contribute to the screened subset, and the same principle extends to screening participants in other perceptual studies.

F BOOTSTRAPPING DETAILS

Because both BBQ and Crowd-BT explicitly model rater quality, we bootstrap over raters rather than over individual comparisons. This approach preserves each rater’s comparison distribution and yields a fair assessment of stability, and it applies uniformly to all of our datasets. Accordingly, we perform 10,000 bootstrap resamples of raters for every dataset, with the sole exception of HUMAINE, where a single bootstrap iteration for Crowd-BT takes roughly 15 minutes, as discussed in Section 4.4. On HUMAINE we therefore use 1,000 resamples to keep computation tractable.

G UNCERTAINTY ESTIMATION DETAILS

Beyond producing a ranking, estimating uncertainty is important in any study for judging whether observed differences are statistically significant.

The Bayesian Bradley-Terry (Bayes-BT) and Bayesian Bradley-Terry with Quality (BBQ) models quantify uncertainty through the posterior distribution over item skills. Each skill carries a Gamma

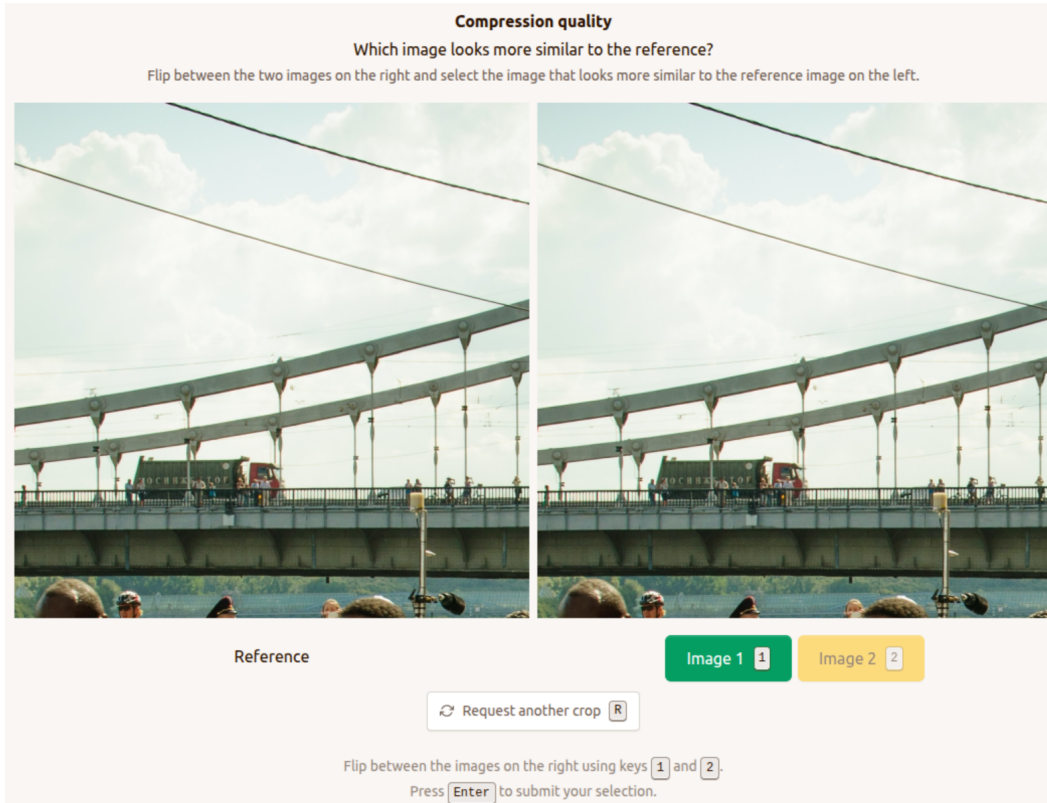


Figure 4: Screenshot of the Mabyduck user study platform used to collect the pairwise comparisons. A reference image is shown on the left, and the rater selects between two compressed images on the right; the same interface generalizes to other pairwise-comparison tasks.

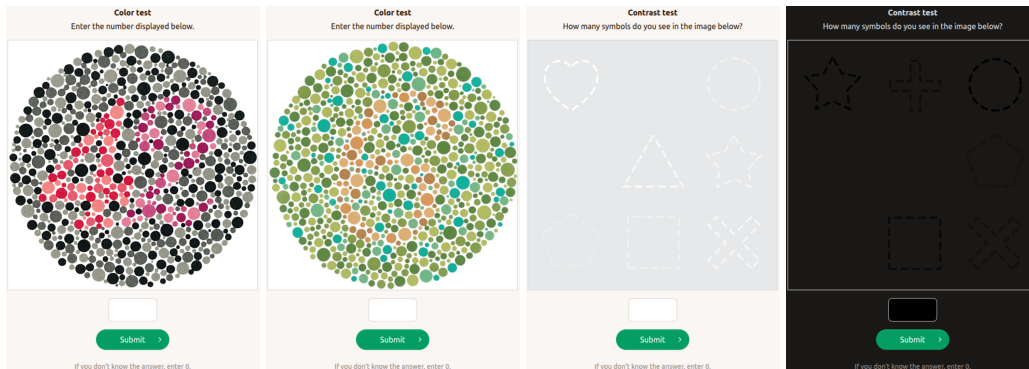


Figure 5: Four example images from the rater pre-screening process. The first two are color blindness tests, in which raters must identify the number displayed in each pattern. The last two are shape detection tests designed to gauge sensitivity to low-contrast objects: one light gray shape on a white background and one dark gray shape on a black background. Screening of this kind is generally applicable to any perceptual comparison study.

prior, which is updated using the observed pairwise comparisons. Credible intervals derived from the posterior are then mapped to the Elo scale to make comparison across items straightforward.

The classical Crowd-BT model, being non-Bayesian, estimates uncertainty through a frequentist approximation. Specifically, it employs a second-order Taylor expansion around the maximum likelihood estimate. The Hessian of the log-likelihood is inverted to obtain the covariance matrix, whose

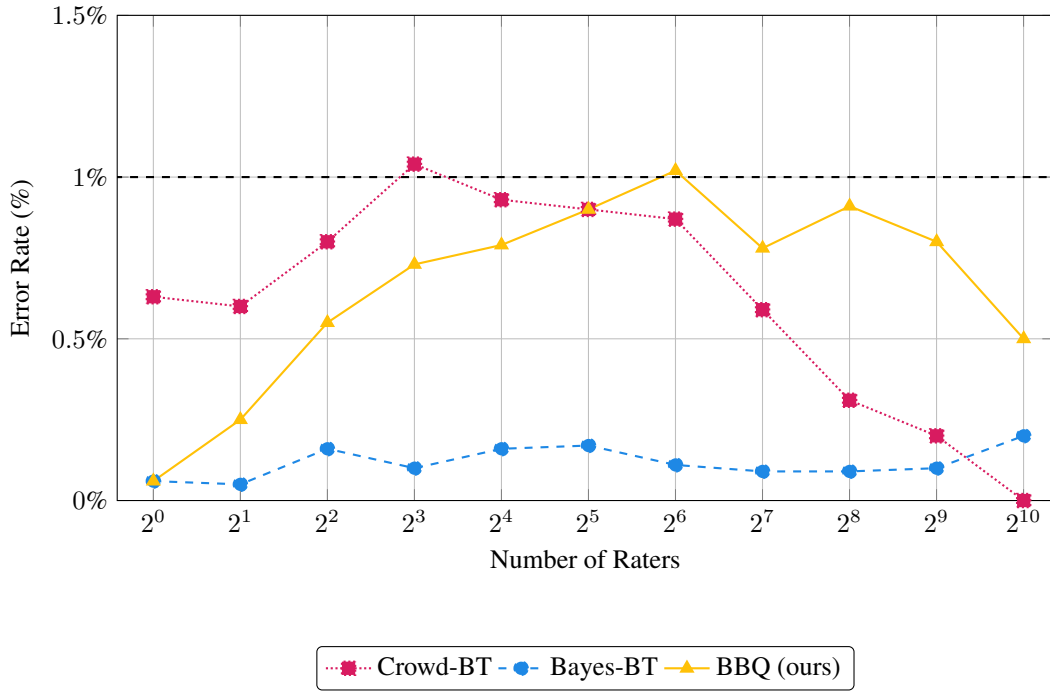


Figure 6: Type I error rate as a function of the number of raters. The dashed line marks the expected error rate of 1%.

diagonal entries give the variances of individual items. These variances are converted to 99% confidence intervals via:

$$p_{99} = \sqrt{\text{diag}(\text{covariance})} \times k_{\text{Erfc}0.01} \times \sqrt{2} \approx \sqrt{\text{diag}(\text{covariance})} \times 3.29,$$

where the constants scale the standard deviation to match the 99% confidence level. Narrower likelihood peaks yield smaller intervals, reflecting higher certainty, while flatter peaks produce larger intervals, indicating greater uncertainty.

To evaluate how well these models estimate uncertainty in a controlled and interpretable way, we run a simulation experiment. We generate trials with two items of equal skill (50–50 win probability) and simulate pairwise comparison data using coin flips. For each trial, we simulate multiple raters (users) and apply the models to estimate Elo scores and their associated uncertainty.

Formally, the null hypothesis H_0 states that the two items are equally strong. For each trial, we compute the 99% confidence (or credible) interval for each item’s skill estimate. A model incorrectly rejects H_0 when the intervals do not overlap, signaling a statistically significant difference between items where none exists. The metric of interest is thus the frequency with which each model wrongly concludes that the items differ, i.e., the observed Type I error rate.

We run 10,000 trials with 50 comparisons per rater, varying the number of raters to trace how the uncertainty estimates scale with the amount of data. This lets us estimate the empirical Type I error rate for each model and compare it against the theoretical expectation of 1% at the 99% confidence level. As shown in Figure 6, Crowd-BT and BBQ exhibit Type I error rates close to the expected 1%, indicating well-calibrated uncertainty estimates. Bayes-BT, by contrast, is overly conservative, consistently producing lower error rates than expected, which suggests its credible intervals are not as well-calibrated for significance testing.